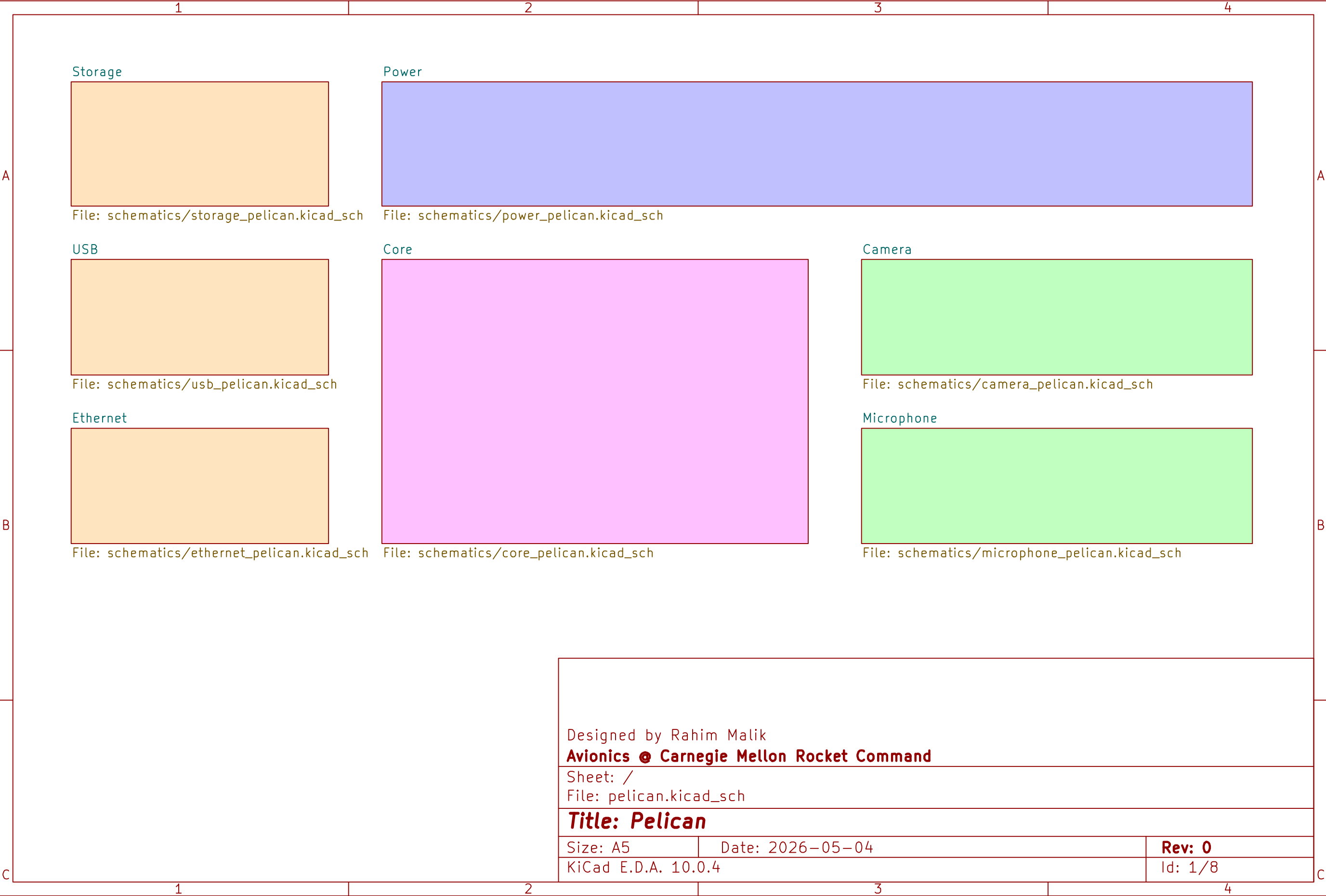




Battery Supply

The diagram illustrates a battery supply circuit for a Raspberry Pi Pico. It includes two battery connectors, J1 and J2, both labeled 436500200. J1 is connected to VBAT and GND. J2 is connected to VCC_3V3 and GND. A switch, SWITCH_STAT, is connected to VCC_3V3 and GND. A pull-up resistor, RPP, is connected to VBAT and GND. The circuit also includes resistors R7 (100R), R8 (100K), R14 (200K), R15 (100K), R16 (100R), R17 (100K), R32 (100K), and R34 (100K). Two MOSFETs, Q1 and Q2, are used for switching. Q1 is a DMG2305UX-13 MOSFET, and Q2 is a DMG2305UX-13 MOSFET. An LED, LED1, is connected to VCC_3V3 and GND. The circuit is powered by VBAT and VCC_3V3, and it controls the VBAT_GATE and VBAT_PROT signals.

Battery Charging

R9 = KISET / ICHG KISET = 890, ICHG = 1A
R12 = KILIM / IIN_LIMIT KISET = 1610, ICHG = 1.5A

The diagram illustrates a battery charging circuit using the BQ24074RGTT IC. The IC is configured with the following components and connections:

- Power Supply:** VCC_3V3 and VCC_5V0.
- Resistors:**
 - R9 (1K10) connected to VCC_5V0 and the IN pin.
 - R12 (887R) connected to VCC_5V0 and the ITERM pin.
 - R11 (100K) connected to VCC_3V3 and the VSS pin.
 - R10 (10K) connected to the BAT pin and GND.
 - R35 (10K) connected to VCC_3V3 and the BAT_CHG_STAT pin.
- Capacitors:**
 - C4 (1uF) connected to VCC_5V0 and GND.
 - C5 (4.7uF) connected to the OUT pin and GND.
 - C6 (4.7uF) connected to the VBAT pin and GND.
- IC Pins and Connections:**
 - IN (13) connected to R9.
 - IILIM (12) connected to R9.
 - TMR (14) connected to R9.
 - ITERM (15) connected to R12.
 - ISET (16) connected to R12.
 - CE (4) connected to GND.
 - EN2 (5) connected to GND.
 - EN1 (6) connected to GND.
 - VSS (8) connected to R11.
 - PAD (17) connected to GND.
 - BAT (2) connected to R10.
 - BAT#2 (3) connected to GND.
 - TS (11) connected to GND.
 - OUT (10) connected to C5.
 - PGOOD (9) connected to GND.
- Status Outputs:**
 - BAT_CHG_SRC_STAT (7) connected to GND.
 - BAT_CHG_STAT (1) connected to R35.
- Other Components:**
 - U3 BQ24074RGTT (IC).
 - U1 (1) connected to VCC_5V0.
 - U2 (2) connected to VCC_5V0.
 - U3 (3) connected to VCC_5V0.
 - U4 (4) connected to VCC_5V0.
 - U5 (5) connected to VCC_5V0.
 - U6 (6) connected to VCC_5V0.
 - U7 (7) connected to VCC_5V0.
 - U8 (8) connected to VCC_5V0.
 - U9 (9) connected to VCC_5V0.
 - U10 (10) connected to VCC_5V0.
 - U11 (11) connected to VCC_5V0.
 - U12 (12) connected to VCC_5V0.
 - U13 (13) connected to VCC_5V0.
 - U14 (14) connected to VCC_5V0.
 - U15 (15) connected to VCC_5V0.
 - U16 (16) connected to VCC_5V0.
 - U17 (17) connected to VCC_5V0.
 - U18 (18) connected to VCC_5V0.
 - U19 (19) connected to VCC_5V0.
 - U20 (20) connected to VCC_5V0.
 - U21 (21) connected to VCC_5V0.
 - U22 (22) connected to VCC_5V0.
 - U23 (23) connected to VCC_5V0.
 - U24 (24) connected to VCC_5V0.
 - U25 (25) connected to VCC_5V0.
 - U26 (26) connected to VCC_5V0.
 - U27 (27) connected to VCC_5V0.
 - U28 (28) connected to VCC_5V0.
 - U29 (29) connected to VCC_5V0.
 - U30 (30) connected to VCC_5V0.
 - U31 (31) connected to VCC_5V0.
 - U32 (32) connected to VCC_5V0.
 - U33 (33) connected to VCC_5V0.
 - U34 (34) connected to VCC_5V0.
 - U35 (35) connected to VCC_5V0.
 - U36 (36) connected to VCC_5V0.
 - U37 (37) connected to VCC_5V0.
 - U38 (38) connected to VCC_5V0.
 - U39 (39) connected to VCC_5V0.
 - U40 (40) connected to VCC_5V0.
 - U41 (41) connected to VCC_5V0.
 - U42 (42) connected to VCC_5V0.
 - U43 (43) connected to VCC_5V0.
 - U44 (44) connected to VCC_5V0.
 - U45 (45) connected to VCC_5V0.
 - U46 (46) connected to VCC_5V0.
 - U47 (47) connected to VCC_5V0.
 - U48 (48) connected to VCC_5V0.
 - U49 (49) connected to VCC_5V0.
 - U50 (50) connected to VCC_5V0.
 - U51 (51) connected to VCC_5V0.
 - U52 (52) connected to VCC_5V0.
 - U53 (53) connected to VCC_5V0.
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 - U78 (78) connected to VCC_5V0.
 - U79 (79) connected to VCC_5V0.
 - U80 (80) connected to VCC_5V0.
 - U81 (81) connected to VCC_5V0.
 - U82 (82) connected to VCC_5V0.
 - U83 (83) connected to VCC_5V0.
 - U84 (84) connected to VCC_5V0.
 - U85 (85) connected to VCC_5V0.
 - U86 (86) connected to VCC_5V0.
 - U87 (87) connected to VCC_5V0.
 - U88 (88) connected to VCC_5V0.
 - U89 (89) connected to VCC_5V0.
 - U90 (90) connected to VCC_5V0.
 - U91 (91) connected to VCC_5V0.
 - U92 (92) connected to VCC_5V0.
 - U93 (93) connected to VCC_5V0.
 - U94 (94) connected to VCC_5V0.
 - U95 (95) connected to VCC_5V0.
 - U96 (96) connected to VCC_5V0.
 - U97 (97) connected to VCC_5V0.
 - U98 (98) connected to VCC_5V0.
 - U99 (99) connected to VCC_5V0.
 - U100 (100) connected to VCC_5V0.

Power Mux

The diagram illustrates a Power Mux circuit using a TPS2116DRLR (U2) to switch between VUSB and VBAT_GATE. The circuit includes resistors R4 (330K), R5 (100K), R6 (10K), capacitors C7, C8, C9, C10, C11, and C12, and a boost converter section with a PWR_SRC input.

Components and Connections:

- U2 (TPS2116DRLR):** The main switching component. Its VIN1 and VIN2 pins are connected to VUSB. Its PR1 pin is connected to VBAT_GATE. Its MODE pin is connected to VUSB. Its VOUT and VOUT pins are connected to the output of the boost converter. Its ST pin is connected to PWR_SRC. Its GND pin is connected to GND.
- Resistors:** R4 (330K) is connected between VUSB and VBAT_GATE. R5 (100K) is connected between VBAT_GATE and GND. R6 (10K) is connected between VCC_3V3 and PWR_SRC.
- Capacitors:** C7 (10uF) and C9 (100nF) are connected between VUSB and GND. C8 (10uF) and C10 (100nF) are connected between VBAT_GATE and GND. C11 (10uF) and C12 (100nF) are connected between the output of the boost converter and GND.
- Boost Converter:** The output of the boost converter is connected to the VOUT and VOUT pins of U2. The boost converter has a PWR_SRC input.

Notes:

- Divider sized to convert USB 4.3V at switch to required 1V threshold

Power Regulation

The diagram illustrates a boost converter circuit for power regulation. The input is labeled BOOST_IN, which is connected to a capacitor C1 (4.7uF) to ground. The input voltage is also connected to the VIN pin (pin 3) of the TPS61023DRLR (U1). The EN pin (pin 2) is connected to the VIN pin through a resistor R1 (100K) and to ground (pin 4) through a green wire. The SW pin (pin 5) is connected to the VIN pin through an inductor L1 (FHD4020S-1R0MT) and to the VOUT pin (pin 6) through a green wire. The VOUT pin is connected to the FB pin (pin 1) through a resistor R2 (768K). The FB pin is also connected to ground through a resistor R3 (100K). The output voltage is labeled VCC_5V0 and is connected to a capacitor C2 (22uF) to ground. The output voltage is also connected to a capacitor C3 (22uF) to ground. A text box provides the formula for the feedback divider: $\text{Divider sized according to } R2 = (V_{\text{out}}/V_{\text{ref}} - 1) * R3$, with $V_{\text{ref}} = 580\text{--}680\text{mV}$, $V_{\text{out}} = 5\text{V}$, and $R3 = 100\text{K}$.

BOOST_IN

C1 4.7uF

GND

R1 100K

GND

U1 TPS61023DRLR

VIN

EN

GND

SW

VOUT

FB

R2 768K

R3 100K

GND

VCC_5V0

C2 22uF

C3 22uF

GND

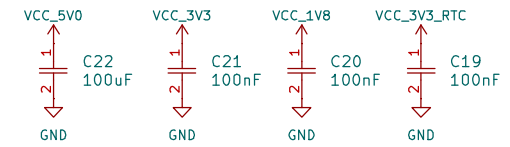
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 $V_{\text{ref}} = 580\text{--}680\text{mV}$, $V_{\text{out}} = 5\text{V}$, $R3 = 100\text{K}$

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Id: 2/8

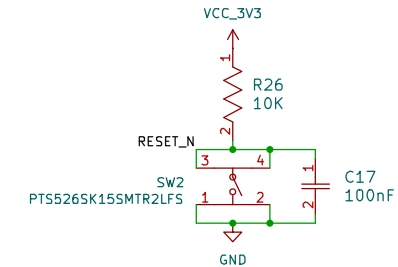
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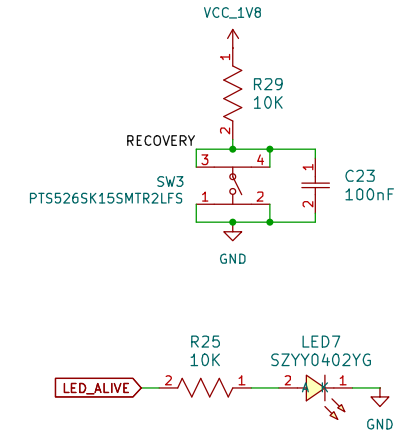
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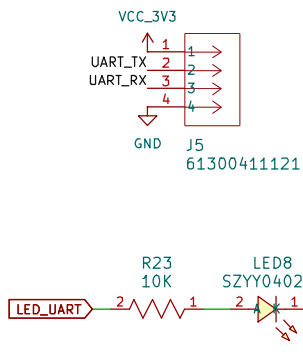
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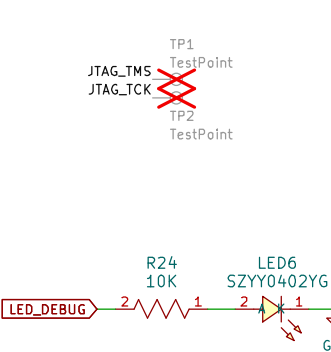
Program



UART



Debug



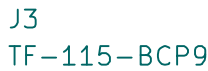
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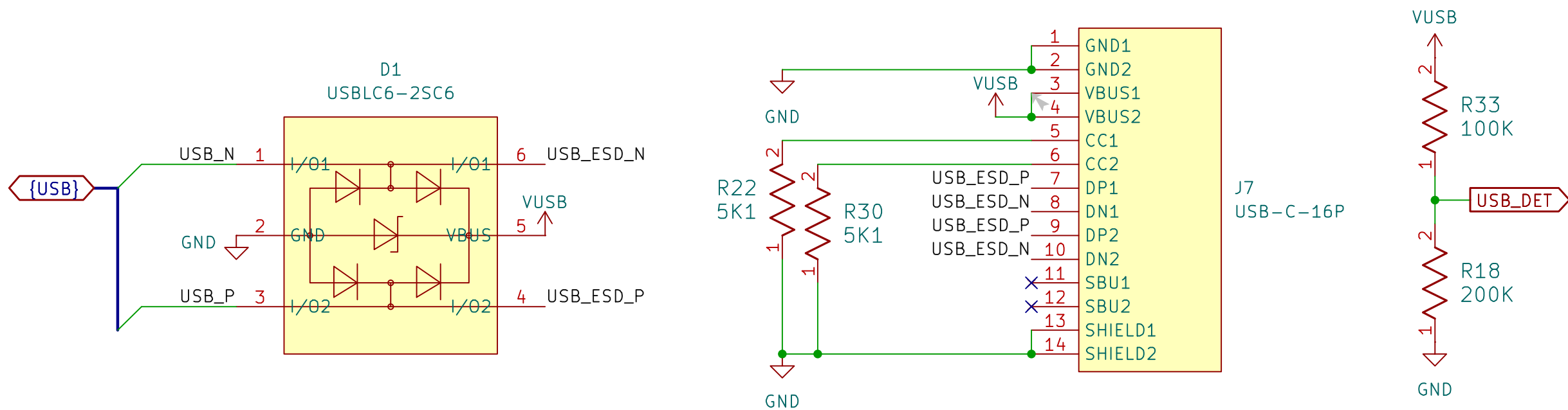
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Date: 2026-05-04

Rev: 0
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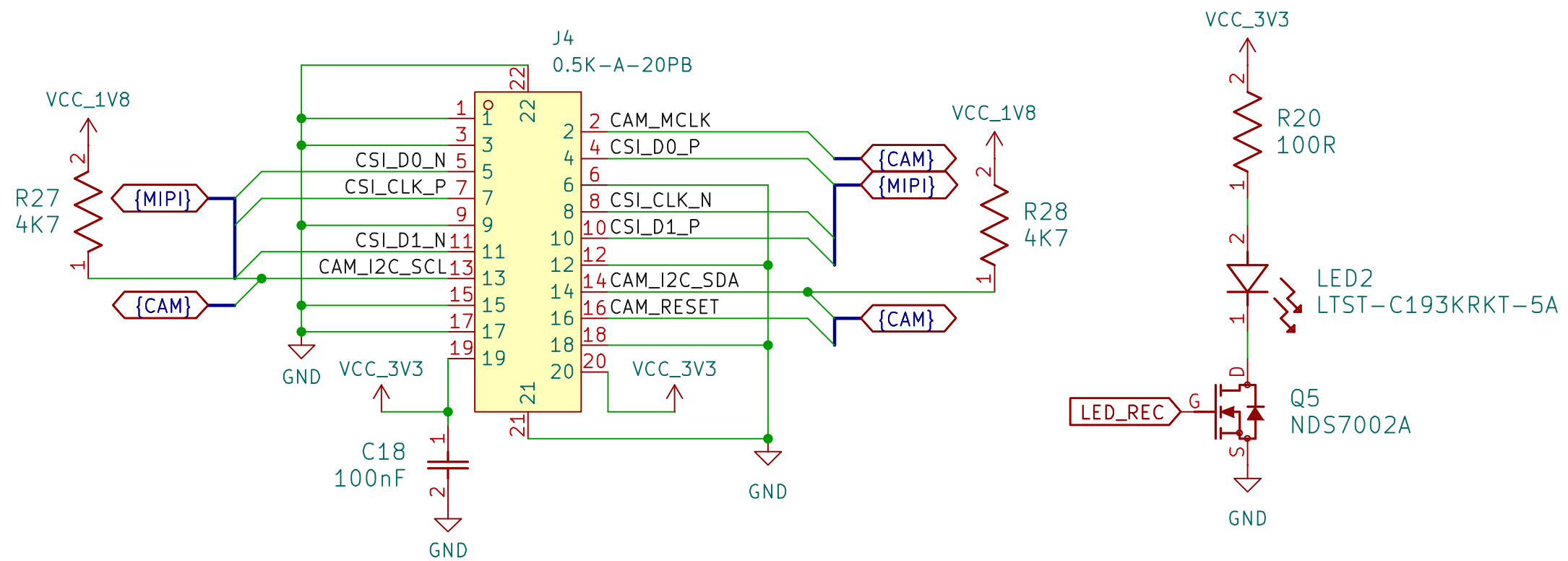
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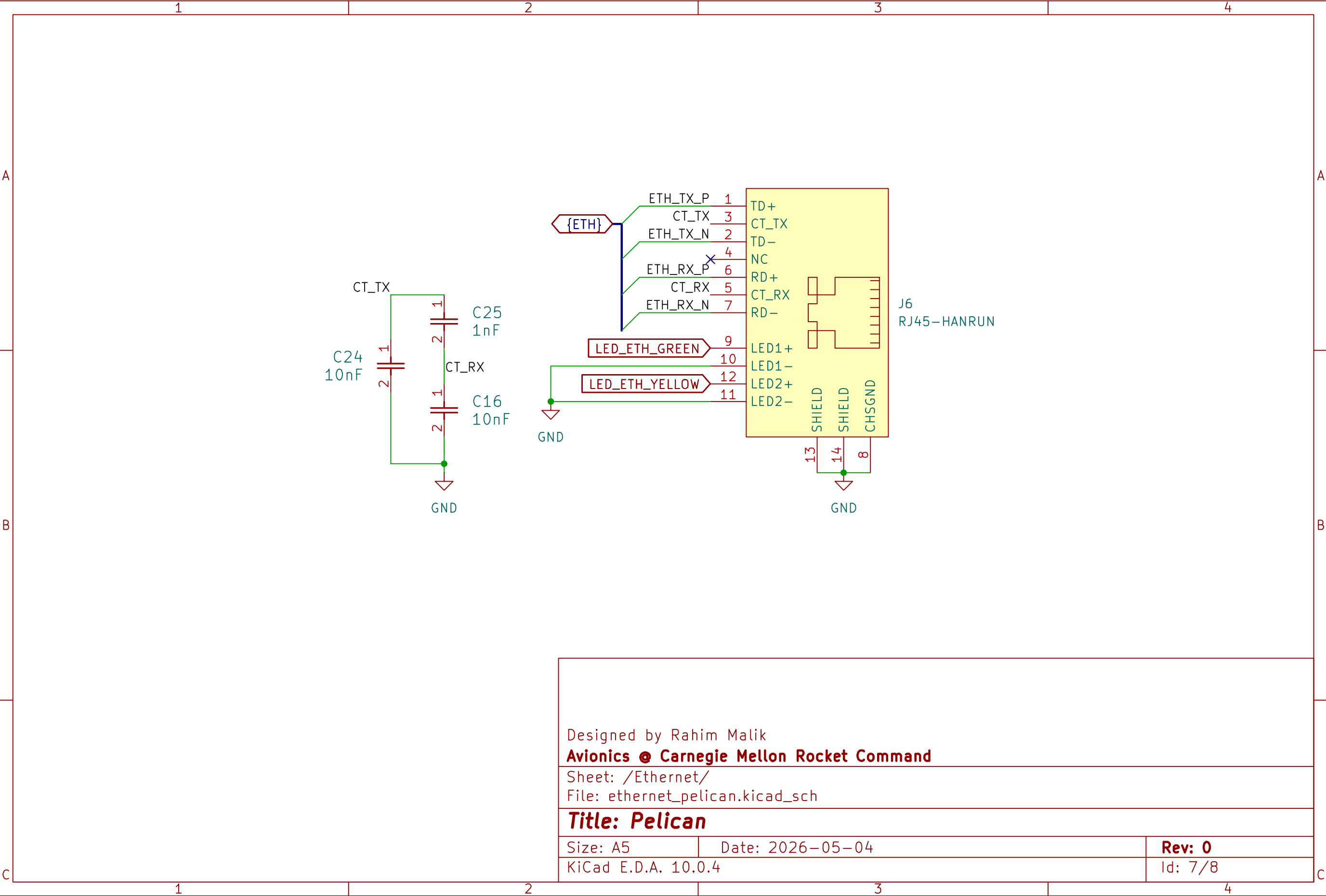
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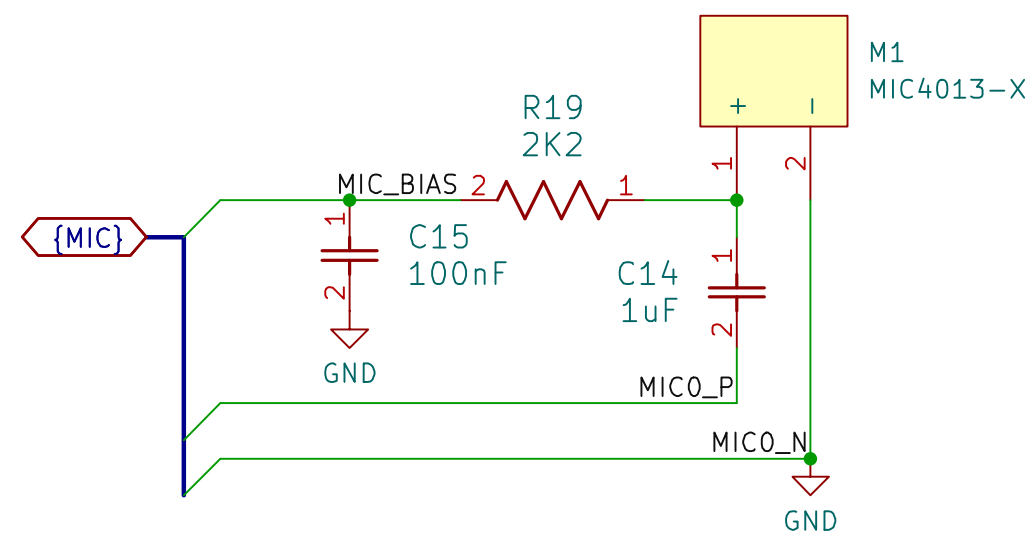
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Size: A5

Date: 2026-05-04

Rev: 0

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